

Nematic Order from Phase Synchronization of Shape Oscillations

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We show that a suspension of noninteracting deformable particles subjected to an oscillatory shear flow leads to development of nematic order that arises from the phenomenon of phase synchronization. The synchronized state corresponds to a unique, stable limit cycle confined in the toroidal state space. The limit cycle exists since, unlike rigid particles, deformable particles can modulate aspect ratio, adjust their tumbling rate, and thus achieve phase synchronization. These synchronized regions emerge as Arnold tongues in the parameter space of the driving amplitude and frequency. Considering the rheological implications of ordering dynamics in soft and active matter, our results motivate oscillatory shear flow experiments with deformable particles.

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Unlike solids and simple liquids, the orientational order of constituent particles is a defining feature of soft and biological matter [1–4]. Classic theories predict the emergence of nematic order, e.g., the Maier-Saupe theory based on anisotropic attractive interactions [5,6] and Onsager’s theory based on the gain in translational entropy at the expense of lowered orientational entropy of the particles [7]. Athermal, active, and external fields [8–12] can also drive the isotropic-nematic transition. Notably, all these approaches rely on interaction between anisotropic particles. They also ignore a characteristic feature ubiquitous in soft and biological matter, namely the deformability of the constituents e.g., as in polymers, soft colloids, emulsions, and cells [13,14]. In this Letter, we show how shape deformability in conjunction with shape anisotropy can induce nematic order of noninteracting particles in a periodically driven system via a phase synchronization mechanism.

It is well-known that two weakly coupled oscillators can synchronize their oscillations [15]. Examples of synchronization are ubiquitous, such as flashing fireflies [16], pendulum clocks [17], mammalian cell cycles [18], beating cilia of microorganisms [19,20], and optomechanical and nanomechanical oscillators [21,22]. A particular case of synchronization is the entrainment of an oscillator by an external, periodic driving. In this case, the coupling is unidirectional, i.e., the external driving can influence the oscillator but not vice versa. Then the frequency and phase

of the driven oscillator gets locked onto the driving signal; and this is known as “phase synchronization” [15,23]. Examples of systems that exhibit phase synchronization due to entrainment are hair cells of the inner ear [24,25], Josephson junctions [26,27], and driven colloidal systems in periodic potentials [28,29].

In this Letter, we consider a system of deformable, tumbling particles dispersed in a simple shear flow as independent oscillators and drive them with a time-dependent, periodic shear. We show that the orientational and shape oscillations of the particles phase-synchronize to the oscillatory shear driving. For many such noninteracting deformable particles, this phase synchronization leads to the emergence of nematic order. As opposed to particle-particle interactions that are traditionally responsible for nematic order in liquid crystals [30,31] or synchronization in Kuramoto models [32], the proposed mechanism is rooted in nonequilibrium driving and is not applicable for rigid particles.

Model—Consider a system of N deformable particles subjected to a time-dependent shear flow $\dot{\gamma}_0[1 + A \sin(\omega t)]$, as schematically shown in Fig. 1(a). The instantaneous state of the i th particle, in two dimensions, is described by two degrees of freedom: (i) a shape parameter r_i equal to the aspect ratio of the particle minus unity, and (ii) an inclination angle θ_i indicating the axis of elongation measured with respect to the flow direction. $r_i = 0$ corresponds to a circular particle whereas larger r_i corresponds to a more elongated particle. The dynamics of the i th particle are described by

$$\frac{dr_i}{dt} = \dot{\gamma}(t)(r_i + 1) \sin 2\theta_i - \frac{1}{\alpha}(r_i - r_0) \left(1 + \epsilon(r_i - r_0)^2\right), \quad (1)$$

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$$\frac{d\theta_i}{dt} = \frac{\dot{\gamma}(t)}{2} (\beta(r_i) \cos 2\theta_i - 1). \quad (2)$$

In Eqs. (1) and (2), $\dot{\gamma}(t) = 1 + A \sin(\omega t)$, $\beta(r_i) = ((r_i + 1)^2 - 1)/((r_i + 1)^2 + 1)$ and time is scaled with the inverse of the mean shear rate $\dot{\gamma}_0^{-1}$.

Equation (1) describes the evolution of the shape parameter [33–35]. The first term on the right-hand side captures the strain-induced deformation of the particle due to velocity gradients in the imposed flow, while the second term is derived from a free energy density, $f_{r_i} = A_r [\frac{1}{2}(r_i - r_0)^2 + (\epsilon/4)(r_i - r_0)^4]$, that maintains an equilibrium particle shape with shape parameter r_0 in the absence of flow. Here, $\alpha = \dot{\gamma}_0/(\Gamma_r A_r)$ controls the strength of free energy induced shape change compared to that of the shear, ϵ describes the quartic free energy landscape, and the parameter Γ_r is a relaxation rate toward the free energy minimum at $r_i = r_0$.

Equation (2) describes the angular velocity of the deformed, anisotropic particle through Jeffery's equation [36]. The two terms in the right-hand-side parentheses, respectively, capture the effect of the strain rate and vorticity components of the shear flow. Therefore, the angular frequency of the tumbling motion depends on the instantaneous shear rate $\dot{\gamma}(t)$, shape parameter $r_i(t)$, and orientation $\theta_i(t)$. The rotational velocity induced by the strain rate (first term) is orientation dependent, and hence, even rigid anisotropic particles ($r_i = \text{constant}$) rotate at a nonuniform rate in a steady shear flow ($A = 0$).

We numerically solve Eqs. (1) and (2) for $N = 200$ particles with random initial orientations (isotropic state). Without loss of generality, we fix $\alpha = 0.8$, $\epsilon = 0.1$, and $r_0 = 5$ in the following analysis.

Emergence of nematic order—Simulations show that, over time, all particles tumble in the flow at the same driving frequency ω , thus frequency locking their rotational motion to the imposed oscillatory shear flow. More importantly, we find that the particles also phase lock their rotational motion with the imposed flow (see Sec. IB of [37]). This phase locking of each particle's orientation with the imposed oscillatory flow implies the synchronization in the motion of all particles with each other as well, and we obtain a state of collective behavior where orientations of all noninteracting deformable particles are the same at all, sufficiently long, times ($t \gg 2\pi/\omega$). Thus, nematic order emerges in the system as indicated schematically in Fig. 1(b) (see Supplemental Video S1 in [37]).

Kymographs illustrating the evolution of the orientation of particles $\theta_i(t)$ are shown in Fig. 1(c) for early and late times. The isotropy in the initial orientation of particles gradually fades away with time, and at long times the phase-synchronized state emerges with tumbling frequency ω and no variations in θ_i between different particles. In contrast, the kymograph obtained for rigid, elongated particles, shown in Fig. 1(d), does not exhibit phase locking

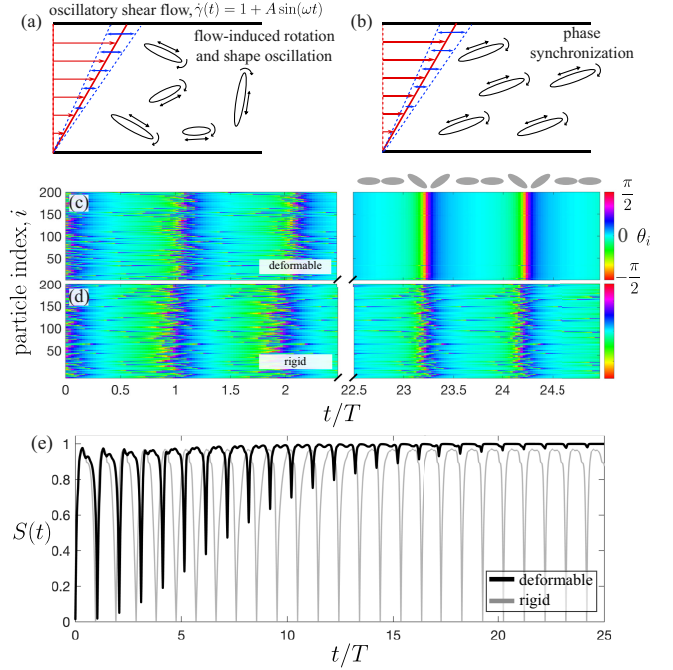


FIG. 1. Emergence of nematic order for noninteracting deformable particles in an oscillatory shear flow. (a) Schematic of the system showing deformable particles suspended in a time-dependent shear flow with both a mean (red) and an oscillatory (blue) component. The initial isotropic state of random orientations $\theta_i(0)$ and varying shape parameters $r_i(0)$ develops into (b) a phase synchronized state at later times. Early and late time kymographs illustrating the temporal evolution of orientation of (c) deformable and (d) rigid particles. (e) The corresponding nematic order parameter $S(t)$ for deformable (black) and rigid (gray) particles. Gray ellipses at top of (c) are a guide for particle orientation while phase synchronized. The plots are for $N = 200$, $A = 0.5$, and $T = 20$.

and the system does not develop nematic order (see Supplemental Video S2 in [37]).

To quantify the nematic order in the system, we calculate the magnitude of the nematic order parameter $S(t) = \langle \cos[2(\theta_i(t) - \Psi_0(t))] \rangle$, where $\langle \cdot \rangle$ denotes an average over N particles and $\Psi_0(t)$ is the mean orientation at time t . The evolution of $S(t)$ for deformable particles is shown in Fig. 1(e) (black curve), which asymptotes toward unity, indicating the development of permanent nematic order in the system. The slow development of nematic order is due to the difference in time taken to phase-synchronize based on initial conditions (see Sec. ID of [37]). Conversely, the measured nematic order parameter in a system of rigid particles subjected to the same oscillatory flow [gray curve in Fig. 1(e)] keeps oscillating between 0 and $\lesssim 1$ [38]. This analysis reveals that phase synchronization, and hence the emergence of nematic order, is a direct consequence of the deformability of particles.

Phase synchronization—We now use a general mathematical framework [15] to demonstrate phase synchronization arising from particle deformability. Representing the

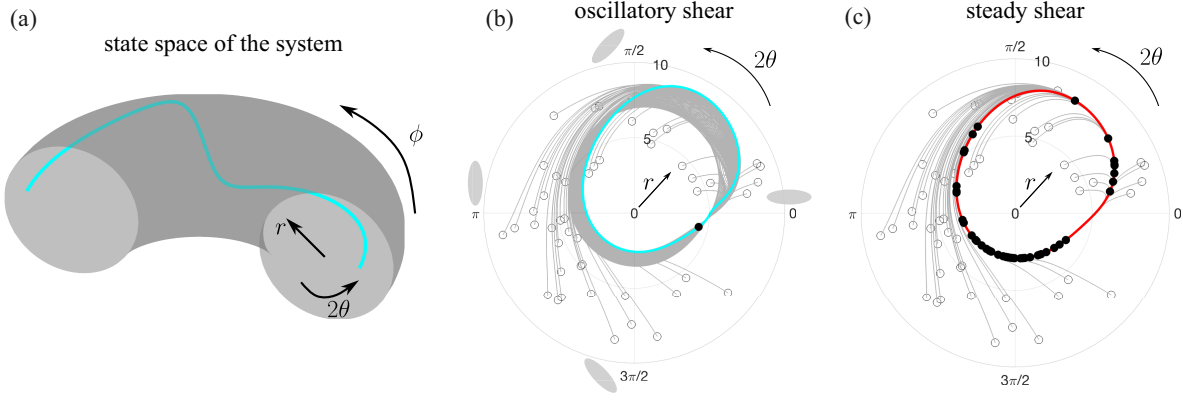


FIG. 2. State space description. (a) Schematic of the 3D state space represented by a torus—driving phase ϕ as the toroidal angle, particle orientation 2θ as the poloidal angle, and particle shape parameter r as the smaller radius. Projection of the torus onto the $(r, 2\theta)$ polar plane for a system subjected to (b) oscillatory shear ($A \neq 0$) and (c) steady shear ($A = 0$). Random initial conditions (open circles) converge to a point (filled circle) on the limit cycle (cyan) in (b), but they converge at different phases on the limit cycle (red) in (c).

dynamical system, Eqs. (1) and (2), in vector form,

$$\dot{\mathbf{x}}_i = F(\mathbf{x}_i; \dot{\gamma}),$$

where $\mathbf{x}_i = (r_i(t), \theta_i(t))$, $F(\cdot)$ is the right-hand-side function, and $\dot{\gamma}(t) = 1 + A \sin(\omega t)$ is the parametric driving. For $A \ll 1$, doing a perturbation expansion up to linear order in A , we obtain

$$\dot{\mathbf{x}}_i \approx F(\mathbf{x}_i)|_{\dot{\gamma}=1} + A \frac{\partial F(\mathbf{x}_i)}{\partial \dot{\gamma}} \Big|_{\dot{\gamma}=1} \sin(\omega t).$$

Performing a phase reduction [15] from the state variables (r, θ) to a phase variable ψ that varies along the oscillation (see Sec. II of [37]), and integrating the resulting equation over one period of the parametric driving $T = 2\pi/\omega$, we obtain the circle map

$$\psi_{k+1} = \psi_k + 2\pi \frac{\omega_n}{\omega} + A f(\psi_k), \quad (3)$$

where ω_n is the natural frequency (i.e., for $A = 0$) and $f(\psi_k) = \int_{kT}^{(k+1)T} (\partial \psi / \partial r_i) (1/\alpha) (r_i - r_0) (1 + \epsilon(r_i - r_0)^2) \times \sin(\omega t) dt$. The map represented by Eq. (3) exhibits phase synchronization in general [39]. On the other hand, for rigid particles $f(\psi_k) = 0$ and they do not phase-synchronize (see Sec. II of [37] for more details). We now turn to investigate the underlying dynamical mechanism responsible for *phase synchronization* of the deformable, tumbling particles.

Dynamical analysis—Equations (1) and (2) constitute a nonautonomous dynamical system due to the time-dependent driving $\dot{\gamma}(t)$ but introducing a new phase variable, $\phi_i = \phi = \omega t$, or

$$\frac{d\phi}{dt} = \omega, \quad (4)$$

with $\phi(t + 2\pi/\omega) = \phi(t)$, results in Eqs. (1), (2), and (4) forming an autonomous 3D dynamical system for a single particle (see Sec. IA of [37] for more details) and allow us to pursue a state-space analysis.

A subset of the complete 3D state space formed by variables r, θ, ϕ for a single deformable particle can be represented as the inside of a torus with ϕ the toroidal angle, 2θ the poloidal angle, and r the smaller radius of the torus [see Fig. 2(a)]. In the synchronized state we find that N different initial conditions corresponding to N particles converge to a stable limit cycle orbit that lies inside the torus (the cyan curve). Furthermore, along the limit cycle all N trajectories converge to the same point at any given time. This is better illustrated in Fig. 2(b), where the azimuthal projection of the torus on to a polar plane formed by $(r, 2\theta)$ is shown. The N trajectories not only converge to the projected limit cycle (cyan curve) but also to the same point (black dot) at a given time indicating phase synchronization.

The focusing of state-space trajectories onto a point on the limit cycle can be understood as follows. All trajectories in state space rotate at the same angular velocity $d\phi/dt = \omega$ around the torus. This implies that N random initial conditions that start at the same initial time [say, $\phi(0) = 0$] must occupy the same instantaneous cross section of the torus—the polar plane formed by r and 2θ —at any future time $\phi(t)$. Since the state-space trajectories must eventually converge onto the stable limit cycle that wraps around the torus, there is only one point along the limit cycle at each cross section, and hence all trajectories will converge toward this point on the limit cycle in each cross section.

This process may be contrasted with a system subjected to a steady shear flow, i.e., $A = 0$. Again, all particles deform and tumble, the trajectories converge to a limit cycle orbit [red curve in Fig. 2(c)] with period T_n but, since the oscillatory shear is absent, the deformable particles occupy different points along the limit cycle at any given

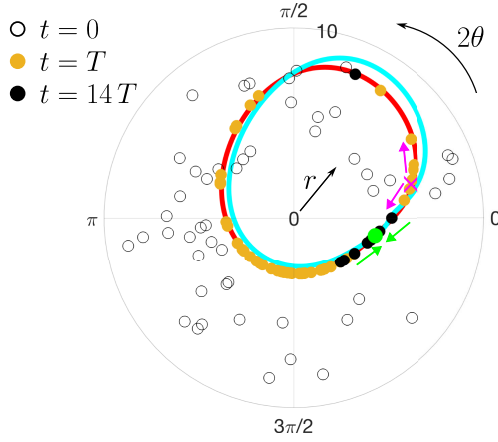


FIG. 3. Transient dynamics in state space. Snapshots of trajectories in the state space as points while passing through the $(r, 2\theta)$ cross section of the torus at three different times. At $t = 0$, random initial conditions (open circles); at $t = T$, trajectories converged to the limit cycle of the steady shear system (yellow filled circles on the red curve), and at $t = 14T$, trajectories (black filled circles) converging onto the stable point (green filled circle with incoming arrows) that lies on the limit cycle of the oscillatory system (cyan curve). The unstable point (pink cross with outgoing arrows) corresponds to the unstable limit cycle in the full state space.

time (black filled circles), i.e., their orientations $\theta_i(t)$ and shape $r_i(t)$ are different. Hence, deformable particles subjected to a steady shear do not phase-synchronize or develop nematic order (see Supplemental Video S3 in [37]).

To further elucidate the transient dynamics of phase synchronization, we plot trajectories on the $(r, 2\theta)$ cross section of the torus, similar to that on a Poincaré section, at three different times: $t = 0$ (black open circles), $t = T$

(yellow filled circles), and $t \gg T$ (black filled circles) in Fig. 3. The initial, randomly distributed points, first, rapidly converge onto the limit cycle of the nonoscillatory system (red curve) and, subsequently, migrate along this limit cycle and converge onto a stable point (green point), which lies on the actual, stable limit cycle of the oscillatory system (cyan curve). In other words, at short times the system behaves effectively like a nonoscillatory system and state-space trajectories rapidly converge onto a “ghost” orbit. At long times, spanning multiple driving periods, the oscillatory driving leads to phase synchronization. Accompanying the stable limit cycle, the unstable limit cycle appears as an unstable point (pink cross) that repels the state-space trajectories in this plot.

Physical mechanism—Figure 4(a) shows the range of amplitude, A , and oscillatory driving period, T , for which phase synchronization occurs. The synchronization regions (violet) occur periodically in bands near multiples of T_n , the natural time period of tumbling in a steady shear flow ($A = 0$). These structures in Fig. 4(a) are the well-known Arnold tongues [15] and they come in and out of existence in parameter space via saddle-node bifurcation of cycles in state space (see Sec. IC of [37]).

The finite width of the tongues can be physically rationalized by analyzing the extent to which a deformable particle can modulate the rate of tumbling and shape changes to match the driving period T of the oscillatory shear. For example, within the primary Arnold tongue in Fig. 4(a), the particle decreases (increases) its tumbling period on the limit cycle when $T < T_n$ ($T > T_n$) by suitable shape deformations. For the tumbling dynamics, since the particle spends most of its time near the $\theta = 0$ ghost region [40] (i.e., region of critical slowdown in state space), the shape modulations near this orientation have the most

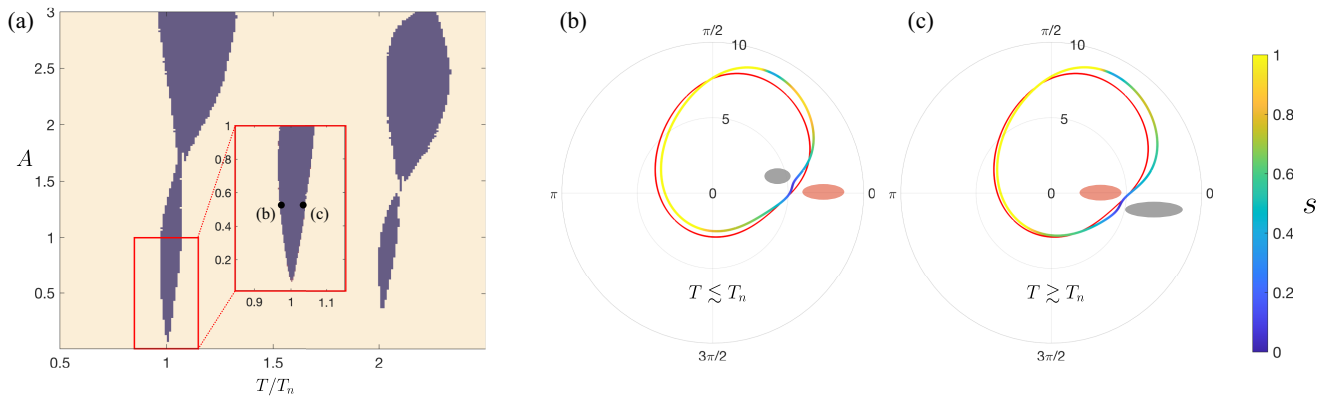


FIG. 4. (a) Arnold tongues: regions of phase synchronization induced nematic order (violet) in the (A, T) parameter space. The inset shows the expanded view of the first tongue. (b), (c) Limit cycles in $(r, 2\theta)$ projection of the state space, corresponding to two points marked in (a): $T = 19.5 \lesssim T_n$ in (b) and $T = 20.7 \gtrsim T_n$ in (c). The color on the limit cycle denotes the state-space speed $s = \sqrt{(dr/dt)^2 + (d\theta/dt)^2}$. In each figure, the limit cycle for the case of steady shear ($T_n \approx 20.02$) is plotted in red. The gray and red ellipses illustrate the aspect ratio of the deformed particle (not to scale) as they pass through the ghost region (near the orientation $\theta = 0$) in the oscillatory and in the steady shear flow, respectively. Compared to particles in steady shear, in (b) particles reduce their anisotropy as they pass through the ghost region to reduce their period. Conversely, in (c) particles increase their anisotropy to increase their period.

significant contribution to the tumbling period. This modulation is illustrated in Figs. 4(b) and 4(c) for $T < T_n$ and $T > T_n$, respectively. In these figures, the limit cycle is colored to denote the state-space speed $s = \sqrt{(dr/dt)^2 + (d\theta/dt)^2}$. The limit cycle for the non-oscillatory system (red curve) is also plotted for comparison. We can see in Fig. 4(b) that compared to that of the nonoscillatory system, the oscillatory system reduces its shape parameter r near the orientation $\theta = 0$. The reduced elongation of the particle helps it traverse the ghost region faster and thus to reduce the period of the limit cycle. Similarly, the limit cycle in Fig. 4(c) has r increased in order to traverse the ghost region slower and thus increase the period of the limit cycle. Hence, by appropriately modulating the shape near the ghost region, the particle matches the period of the limit cycle with the imposed oscillation period.

Discussion and outlook—We have shown that deformable anisotropic particles can phase-synchronize in time-dependent oscillatory shear flow. At the collective level, this results in the development of nematic order in a suspension of noninteracting particles. For $A \gg 1$, the mean shear flow component is superseded by the oscillatory component, and the particles undergo complex yet synchronized back-and-forth angular oscillations and tumbling motion (see Secs. IE and IF of [37]). Our results also extend to continuum models of deformable particles by modifying $\beta(r)$, the flow aligning scale (see Sec. V of [37]). Furthermore, the phenomenon of synchronization-induced nematic order occurs in steady shear flow if the noninteracting particles are active. In Secs. III and IV of [37], we provide a mathematical analysis and numerical results to show phase synchronization for particles that actively undergo shape oscillations or have periodically varying chiral activity, respectively. Such situations may be relevant for active particles undergoing shape deformations [41–44] or in general when the active particles are undergoing periodic modulations in an internal state. Hence, our Letter opens up new avenues to explore at the intersection of nonlinear dynamics of synchronization and soft and active matter.

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Data availability—The data that support the findings of this Letter are not publicly available upon publication

because it is not technically feasible and/or the cost of preparing, depositing, and hosting the data would be prohibitive within the terms of this research project. The data are available from the authors upon reasonable request.

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- [37] See Supplemental Material at <http://link.aps.org/supplemental/10.1103/j19g-c2nh> for additional results of phase synchronization of deformable particles (Sec. I), a mathematical analysis of synchronization for deformable and rigid particles (Sec. II), synchronization of elongated particles with active shape deformations (Sec. III), synchronization of rigid elongated particles with periodically varying chiral activity (Sec. IV), and synchronization in continuum model of deformable nematic particles (Sec. V), which includes Supplemental Videos S1–S5.
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